Velagapudi Mahendra

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SUMMARY

Data science and machine learning student (B.Tech, 2023 – 2027) who builds and rigorously evaluates end-to-end systems: low-resource Telugu NLP with significance testing, an automated air-quality data pipeline, a citation-grounded RAG chatbot, and reinforcement-learning reward experiments. Seeking Data Science, ML or Data Analyst internships and entry-level roles.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, HTML, CSS, JavaScript, TypeScript
ML and Data: PyTorch, Hugging Face Transformers, scikit-learn, Stable-Baselines3, Pandas, NumPy, Jupyter
Data Engineering: SQLAlchemy, PostgreSQL, SQLite, ChromaDB, GitHub Actions, REST APIs, Docker
Tools and Platforms: Streamlit, Plotly, Matplotlib, Power BI, Git, VS Code, Linux, pytest

PROJECTS

Telugu Text Classifier | *Python*, *PyTorch*, *Transformers*, *scikit-learn*, *Gradio* GitHub

- Fine-tuned IndicBERTv2, MuRIL and XLM-RoBERTa on 35,142 Telugu sentences (3-class sentiment); IndicBERTv2 reached macro-F1 0.733 vs 0.648 for a TF-IDF baseline (+8.5 points, $p < 0.001$, paired bootstrap).
- Removed 143 exact and 538 near-duplicates before splitting to prevent leakage; labeled 100 errors into a 5-category taxonomy; shipped a Gradio demo.

Hyderabad Air Quality Monitor | *Python*, *SQLAlchemy*, *PostgreSQL*, *GitHub Actions*, *Streamlit* GitHub | Live demo

- Built a daily automated pipeline ingesting OpenAQ data from 21 stations and 273 sensors into PostgreSQL with idempotent upserts: 10,262 records, zero rejected, 3 of 3 scheduled runs successful.
- Added rolling-MAD anomaly detection (148 readings flagged), an interactive Streamlit dashboard and 27 passing tests.

Messy PDF RAG Chatbot | *Python*, *PyMuPDF*, *ChromaDB*, *ONNX embeddings*, *Streamlit* GitHub

- Built a RAG system over 18 technical PDFs (535 pages, 1,629 chunks) with OCR fallback, page-aware chunking and verified page citations.
- Evaluated on a 66-question golden set: Hit Rate@5 0.926 and MRR 0.893; 174 passing tests.

Lunar Lander Reward Shaping Lab | *Python*, *Gymnasium*, *Stable-Baselines3*, *PyTorch* GitHub

- Trained PPO agents (3 seeds, 1M steps each) under four reward designs, evaluated on the unshaped reward over 300 episodes: a hover bonus caused reward hacking (85% timeouts vs 1% baseline) while potential-based shaping kept a 97% landing rate.
- Found and fixed a bug where the reward-mode flag never reached the wrapper; added regression tests (51 passing).

EXPERIENCE

Data Science Intern, Codec Technologies Pvt. Ltd (Remote) *May 2025 – Jun 2025*

- Contributed to VectorMind, a sales-analytics platform designed to explain trends, forecast sales, and answer natural-language questions about sales data.

ACTIVITIES

Google Tech Sprint Hackathon, 4th place *2025*

- Built MDM, a certificate issuance and verification platform for institutions, students and employers: cryptographically signed tamper-evident credentials, automated PDF generation, IPFS anchoring, instant QR verification and role-based access control.

EDUCATION

B.Tech, Computer Science and Engineering (Data Science) *2023 – 2027*
Malla Reddy Engineering College & Management Sciences, Hyderabad | 80% (through 6th semester)

CERTIFICATIONS

Data Analytics and Visualization Job Simulation, Forage (Aug 2025) | Data Analytics Job Simulation, Deloitte (Aug 2025)
SQL Bootcamp, Let's Upgrade (Aug 2025) | Data Science Program, Micro IT (Aug 2025)
Languages: English (Professional), Telugu (Native), Japanese (Beginner)